Dependent Gaussian Spectrum of Staple Dispersion

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Abstract

We compute the full projected-Gaussian quadratic spectrum of linearized staple dispersion on the open 2^4 block and prove that the low-dispersion conditional window overlaps a subcritical high-dispersion polymer window.

Let P be the scalar curvature projection. For a six-plaquette link star with selector S and centering matrix $C = I - \mathbf{1}\mathbf{1}^T/6$, the linearized dispersion is the three-color quadratic form with covariance spectrum

$$\operatorname{spec}_+(CSPS^TC).$$

Theorem 1. *All eight six-staple interior links have the same five nonzero eigenvalues, printed by the executable certificate, each at most one. For threshold D and heat time t,*

$$\Pr(Q \geq D) \leq e^{-I(D)}, \quad I(D) = \sup_{0<\lambda<(2t\lambda_{\max})^{-1}} \left[\lambda D + \frac{3}{2} \sum_j \log(1 - 2\lambda t \lambda_j) \right].$$

No plaquette independence is assumed.

Proof. Centering cancels the selected-link linear variable. The global curvature covariance is $tP \otimes I_3$. The Gaussian determinant formula gives the moment generating function and Chernoff optimization. Exact finite incidence symmetry yields the common spectrum. □

Multiplying the tail by degree-18 connected entropy produces a rooted free energy. Its inverted lower RMS threshold is strictly below the maximum RMS dispersion accepted by the coherent conditional estimate. Thus the dependent Gaussian model has a nonempty common dispersion interval.

Remark 1. *This closes the split only in the exact linearized Gaussian block. A retained-state-uniform nonlinear density/repair comparison is still required.*